

AI-Assisted House Price Estimation

End-of-Course project

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Repository: <https://github.com/xjj2109454658-wq/PE6201-end-of-course-project.git>

Video: <https://youtu.be/Z1poN8rGQak>

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Section 1 · Project Overview and Design Rationale

Our project, AI-Assisted House Price Estimation, estimates residential sale prices from property characteristics. Our initial design compared Linear Regression and Random Forest using structured numerical inputs. However, feedback highlighted that this did not demonstrate the role of AI beyond conventional machine learning.

We therefore redesigned the system as a hybrid AI application. Figure 1 shows the final architecture and information flow.

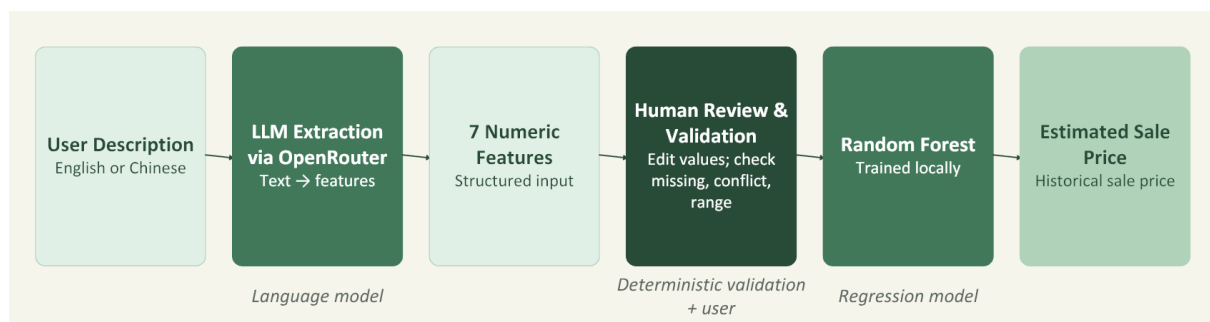


Figure 1. Final system architecture showing the flow from free-text input to price estimation

Instead of entering seven numerical variables manually, users can describe a property naturally in English or Chinese. An OpenRouter-hosted language model extracts seven required features, which are shown for user review and correction before prediction. The Random Forest model then generates the estimated sale price.

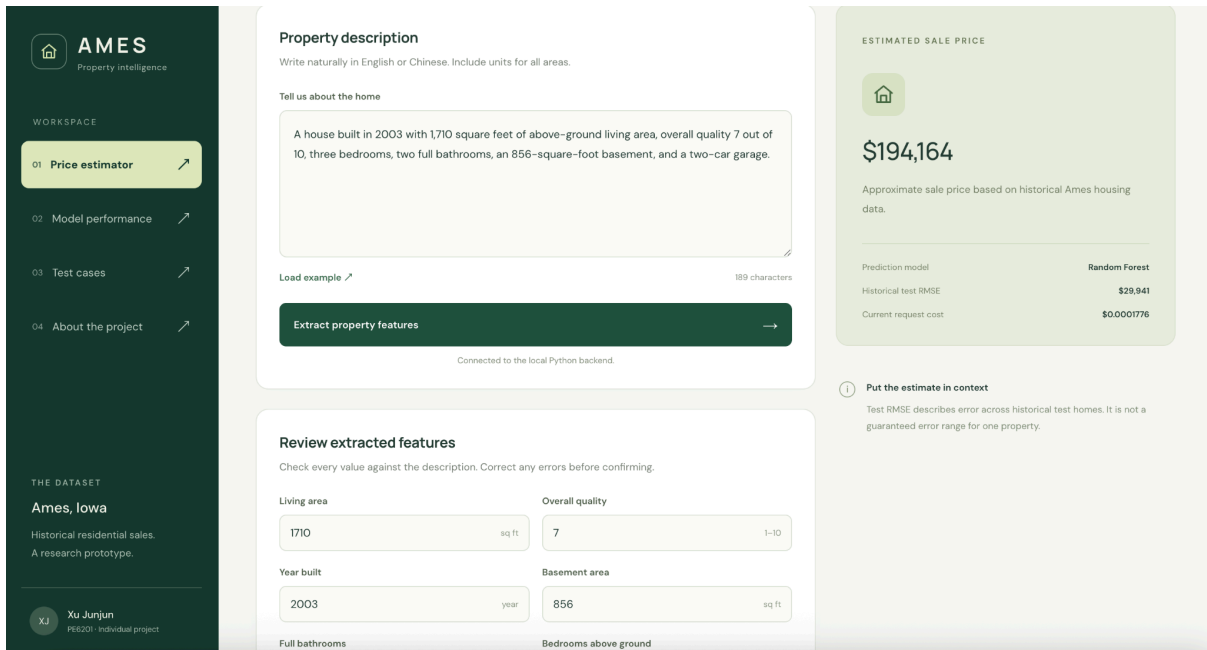


Figure 2. Price Estimator interface

This separation was deliberate: the LLM handles language understanding, while the regression model performs numerical prediction. We avoided asking the LLM to predict price directly because a plausible-looking output may not reflect the statistical relationships learned from our dataset. Keeping prediction within the trained regression model makes the core task more reproducible and easier to evaluate.

Section 2 · Data and Model Performance

The system uses the Ames Housing dataset, containing 1,460 labelled homes. We used 1,168 homes for training and 292 held-out homes for testing, with the same seven features used for both Linear Regression and Random Forest. The mean sale price in the test set was approximately \$178,840, which provides necessary context for interpreting the error. Random Forest achieved a test RMSE of \$29,941, compared with \$38,999 for Linear Regression. This represents a 23.2% lower RMSE for Random Forest, so we selected it for the final prototype.

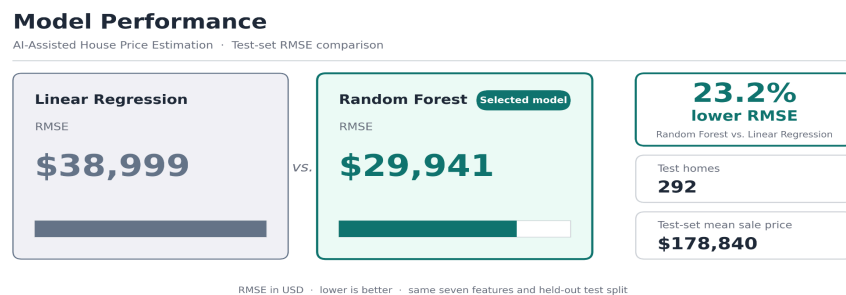


Figure 3. Model Performance interface comparing Linear Regression and Random Forest.

However, this result should not be interpreted as evidence that the Random Forest prediction will normally be within \$29,941 of the true price. RMSE is an aggregate measure across the test set rather than an error bound for an individual property. We therefore explicitly communicate this distinction in the interface. This is important because a house-price application has a particularly strong risk of creating false confidence: a precise-looking dollar amount can appear more authoritative than the underlying evidence justifies.

The comparison also demonstrates why model selection should follow evidence rather than complexity. Random Forest is more complex than Linear Regression, but in our test it produced substantially lower error using exactly the same features and test split. Therefore, the additional complexity was justified by measurable improvement rather than by assuming that a more sophisticated model must be better.

Section 3 · Evaluation and What It Actually Proves

We evaluated the complete application through six scenarios rather than testing only successful predictions. The first two tested valid English inputs, including a larger home. The third used missing information, the fourth contained conflicting garage information, the fifth supplied a construction year outside the training range, and the sixth tested Chinese input with square metres. The valid cases produced estimates such as \$194,164, \$332,000, and \$195,468, while the three problematic cases were blocked for review.

Extraction and validation checks

Six end-to-end checks verified through the connected frontend and backend.

CASE	SCENARIO	STATUS	RESULT
01	English · square feet	Complete	\$194,164
02	Larger valid home	Complete	\$332,000
03	Missing required fields	Blocked	OverallQual, TotalBsmtSF, GarageCars
04	Conflicting garage capacity	Blocked	GarageCars requires clarification
05	Year outside training range	Blocked	2022 outside [1872, 2010]
06	Chinese · square metres	Complete	\$195,468

VERIFIED TEST CASES

6

These checks confirm specific extraction, validation and prediction behaviours; they do not establish general model or extraction accuracy.

Figure 4. The validation layer distinguishes valid and problematic inputs.

The evaluation tests more than prediction accuracy: it tests whether the system knows when not to predict. Missing, conflicting, or out-of-range values are blocked by deterministic validation rather than passed to the regression model.

However, six scenarios cannot establish general extraction accuracy. An incorrectly extracted value may still appear numerically valid. For example, if the LLM misunderstands a property's living area but returns a plausible number, validation may not detect the error. This creates a potential silent failure, where an incorrectly interpreted input still produces a plausible-looking price.

The system therefore requires human review before prediction. A stronger evaluation would use a larger, systematically constructed dataset of property descriptions with ground-truth features to measure extraction and prediction accuracy separately, rather than treating the whole pipeline as a single black box.

Section 4 · Trade-offs, Cost and System Design Decisions

A major change from the original design was the introduction of an LLM extraction layer. This creates several important trade-offs rather than simply making the system “more intelligent”.

The first is convenience versus cost. Free-text input is easier for users because they do not need to understand the seven-field input schema. However, each estimate requires an OpenRouter API call, which introduces an additional usage cost that would not exist in a purely local regression system. The cost per estimate can therefore be viewed as the price of providing a more flexible user experience. Although the API cost of a single extraction is small in a prototype, it becomes more relevant as usage scales.

The second is flexibility versus reliability. An LLM can interpret natural descriptions and handle different wording or units, including Chinese descriptions and square metres. However, this flexibility introduces the possibility of semantic extraction errors. A traditional structured form is less convenient but makes the expected input explicit. We therefore accepted some additional AI risk in exchange for a more natural interface.

The third is automation versus human oversight. Fully automatic extraction would provide the fastest experience, but an incorrect feature could be passed directly to the regression model. Our design adds a human review step, which increases interaction time but reduces the risk of an unnoticed extraction error. This is an intentional trade-off between convenience and reliability.

There is also a prompting trade-off. More detailed prompts can constrain the LLM to return the required features and format, but longer prompts may increase token usage and therefore cost. We therefore use prompting to define a narrow extraction task rather than asking the model to perform the entire valuation. This keeps the LLM's role focused and makes its behaviour easier to evaluate.

These trade-offs also reflect the broader rent-versus-build consideration raised during the project feedback. Using an external hosted LLM means relying on an API rather than building and maintaining our own language-understanding component. The API approach reduces development complexity and provides access to a capable language model, but introduces ongoing usage cost and

dependency on an external service. For a small prototype, this trade-off is reasonable; at larger scale, cost, latency, privacy, and service dependency would need further evaluation.

Section 5 · How the Project Changes the Current Experience

Compared with a conventional house-price model, our system changes the interaction rather than attempting to solve every aspect of valuation. Users can describe properties naturally, including Chinese input and common units such as square metres, while the system converts the description into the structured representation required by the model.

More importantly, the review step makes extraction uncertainty visible. The key contribution is therefore not simply using AI to predict prices, but combining technologies according to their strengths and limitations.

Section 6 · Rough Edges and Future Work

The prototype still has several limitations. First, the Ames Housing dataset is historical and limited to homes in Ames, Iowa; its newest construction year is 2010. The model should therefore not be treated as a current market valuation tool or directly applied to other markets without local data. Second, only seven numerical features are used, excluding potentially important factors such as neighbourhood. Third, the LLM can produce semantically incorrect but numerically valid values, so human review remains necessary. Finally, the model retrains at startup rather than loading a saved model, which is acceptable for a prototype but inefficient for production.

Future work should expand the feature set, systematically measure extraction accuracy, and evaluate additional regression models. A larger test set could also identify which types of descriptions cause the most extraction errors. At deployment scale, API usage, latency, and cost should be monitored.

Section 7 · Conclusion

The main lesson from this project is that a stronger AI system is not necessarily one that gives the AI more control. Our final architecture assigns different responsibilities to different components: the LLM provides language understanding, deterministic validation enforces constraints, human review handles uncertainty, and Random Forest performs numerical prediction.

The project also shows that GenAI introduces genuine trade-offs. Natural-language interaction improves convenience, but adds API cost and new failure modes. Human review reduces automation but provides protection against silent extraction errors. The prototype therefore succeeds not simply because it produces a house-price estimate, but because the path to that estimate is explicit and reviewable. Its limitations also prevent us from treating it as a professional valuation system.